Avneet Aurora

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EDUCATION

University of Delaware, Newark, DE **B.S. Computer Engineering – Honors College** | GPA: 3.77 | **Dean's List**
Coursework: Calculus III, Linear Algebra II, Data Structures & Algorithms, Software Engineering, Machine Learning, Digital Logic Design

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, Java, HTML, CSS, VHDL
ML & Data: PyTorch, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib
Backend & Cloud: FastAPI, Node.js, REST API design, PostgreSQL, AWS (EC2, RDS), CI/CD, Linux/Bash, Git
Frontend & Mobile: React, Next.js, React Native (Expo), HTML/CSS

RELATED EXPERIENCE

Undergraduate Research Assistant — Vertically Integrated Projects (Scientific Computing) *Feb 2026 – Present*

- Extended peer-reviewed 2D Lattice Boltzmann fluid-simulation research to 3D (D3Q27 lattice) by replacing the analytical BGK collision operator with a learned neural network, enabling data-driven simulation of turbulent flows
- Improved model generalization and shrank the parameter search space by engineering TensorFlow/NumPy training pipelines that enforce physical lattice-symmetry constraints on the learned operator
- Achieved mass-conservation error at machine precision (~1e-15) by validating the 3D model against analytical solutions, confirming the operator preserves the underlying physics; presented results to the research team

Undergraduate Research Assistant — Winter Fellows, Electrical & Computer Engineering *Jul 2025 –Feb 2026*

- Raised wildfire-ignition validation accuracy by 10–20% over a baseline stuck at ~50% by re-architecting a PyTorch CNN (ReLU, batch normalization, regression head), correcting the data pipeline, and expanding the training set
- Selected for the merit-based Winter Fellows award to extend 3D ignition-estimation research from airborne to satellite LiDAR point clouds; presented findings at the Winter Undergraduate Research Showcase
- Reduced per-plot preprocessing from a manual multi-step workflow to a single scripted run by automating Python pipelines for point-cloud normalization, forest-metric computation, and heatmap visualization

Technology Intern — Code Differently – Buccini Pollin Group, Wilmington, DE *Jun 2024 – Aug 2024*

- Performed financial analysis for real estate development projects; applied problem-solving and time management skills to present a data-driven proposal to BPG founders on amenity gym improvements.
- Earned Microsoft Office, Excel, Word, and Digital Literacy certifications; awarded a $1,000 Metropolitan Urban League scholarship

PROJECT EXPERIENCE

Decora AI— decora.club *Aug 2025 – Nov 2025*

- Built a full-stack AI interior-design platform using React/Next.js + TypeScript; designed RESTful API endpoints (API design) for serverless backend deployment on Vercel. Learned agile methodologies in software engineering and pushed production-level code.
- Implemented backend development features, including Supabase Auth/DB/Storage integration and a Stripe-based credit system with object-oriented design patterns. Used SQL databases to store user information and placed images in buckets.
- Applied, tested, and analyzed data modeling for furniture detection pipelines and user session management; optimized user experience (UX) across the end-to-end redesign workflow

TaskAI – https://devpost.com/software/taskai-vjx8cl *Dec 2025*

- Developed a Python web app for college students to schedule study sessions around classes and activities; integrated Google Gemini for intelligent scheduling recommendations
- Designed a frontend with a clean user interface (UI) in Drafter with a focus on user experience (UX); won Best Functional Implementation at the University of Delaware Honors Hackathon.

SquatBuddy — devpost.com/squatbuddy *March 2026*

- Under biotech and healthcare, built an AI-powered powerlifting coach at HenHacks 2026 using an LSTM Autoencoder to analyze squat form frame-by-frame from raw video input.
- Applied deep learning and object-oriented design from OpenCV libraries to deliver joint-specific biomechanical feedback; engineered a real-time video processing pipeline across the full kinetic chain.

LEADERSHIP & AWARDS

DevLab Club — President, Newark, DE *Feb 2025 – Present*

- Led and organized weekly sessions on full-stack and backend development topics, including Git, API design, Software Engineering Principles like object-oriented programming, and use of LLMs like Gemini or ChatGPT; grew club to 20+ active members
- Mentor and collaborate with members, using verbal communication skills to outline personal project design, problem-solving, and debugging code and architecture.

Winter Fellows Award, *Dec 2025*

Best Functional Implementation Award, *Dec 2025*